

German Barquero

Curriculum Vitae



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Google Scholar for Full List of Publications

RESEARCH EXPERIENCE

Meta Superintelligence Labs *AI research scientist*

JUN 2025 - NOW (FT)

Developing real-time interactive video world models.

Meta Reality Labs *Research scientist intern*

JUN 2024 - NOV 2024 (FT - 6 MONTHS)

Led the research and development of RPM, a real-time full-body motion generation model for XR. First author of the resulting CVPR 2025 paper and co-creator of the GORP dataset.

Herta Security *Applied research scientist*

SEP 2020 - MAR 2021 (PT - 7 MONTHS)

I developed a very fast and accurate face tracking algorithm (T-BIOM 2021) and worked on quantization and object detection.

École polytechnique fédérale de Lausanne + CHUV *Research scientist intern*

SEP 2019 - AUG 2020 (FT - 1 YEAR)

In collaboration with an interdisciplinary medical imaging team, I explored the capabilities of several MRI modalities for detecting paramagnetic rim lesions in multiple sclerosis (RimNet, *NeuroImage: Clinical*) and the challenges of ultra-high-field MRI.

Herta Security *Research scientist intern*

JUN 2019 - SEP 2019 (FT - 4 MONTHS)

I conducted research on efficient tracking algorithms. I collected and annotated a dataset for face tracking in large crowds and developed a fast novel face tracking algorithm (awarded at IJCB 2020).

EDUCATION

2022 – 2025 **PhD in Mathematics and Computer Science**
SUMMER SCHOOLS: ICVSS'23, GEMSS'23
Universitat de Barcelona

2020 – 2021 **Master in Computer Vision**
BEST STUDENT AWARD (1/30)
Universitat Autònoma de Barcelona

2014 – 2020 **Bachelor of Mathematics**
Bachelor of Computer Science
Universitat de Barcelona

SELECTED PUBLICATIONS

Hong, C., **Barquero, G.**, et al. (2025). SneakPeek: Future-Guided Instructional Streaming Video Generation. *arXiv 2025*.

Barquero, G., et al. (2025). From Sparse Signal to Smooth Motion: Real-Time Motion Generation with Rolling Prediction Models. *CVPR 2025*.

Ruiz-Ponce, P., **Barquero, G.**, et al. (2025). MixerMDM: Learnable Mixing of Human Motion Diffusion Models. *CVPR 2025*.

Barquero, G., Escalera, S., and Palmero, C. (2024). Seamless Human Motion Composition with Blended Positional Encodings. *CVPR 2024* (>250 ⭐).

Ruiz-Ponce, P., **Barquero, G.**, et al. (2024). in2IN: Leveraging individual Information to Generate Human INTERactions. *CVPR Workshops 2024*.

Song, S., Spitale, M., Luo, C., Palmero, C., **Barquero, G.**, et al. (2024). REACT 2024: the Second Multiple Appropriate Facial Reaction Generation Challenge. *FG Challenge 2024*.

Barquero, G., Escalera, S., and Palmero, C. (2023). BeLFusion: Latent Diffusion for Behavior-Driven Human Motion Prediction. *ICCV 2023* (>100 ⭐).

Song, S., Spitale, M., Luo, C., **Barquero, G.**, et al. (2023). REACT2023: The First Multiple Appropriate Facial Reaction Generation Challenge. *ACM MM Challenge 2023*.

Barquero, G., et al. (2022). Didn't see that coming: a survey on non-verbal social human behavior forecasting. *ICCV Workshops 2021*.

Barquero, G., et al. (2022). Comparison of Spatio-Temporal Models for Human Motion and Pose Forecasting in Face-to-Face Interaction Scenarios. *ICCV Workshops 2021*.

Palmero, C., **Barquero, G.**, et al. (2022). Chalearn LAP challenges on self-reported personality recognition and non-verbal behavior forecasting during social dyadic interactions: Dataset, design, and results. *ICCV Workshops 2021*.

Barquero, G., Hupont, I., and Fernandez, C. (2021). Rank-based verification for long-term face tracking in crowded scenes. *IEEE T-BIOM*.

Barquero, G., Fernandez, C., and Hupont, I. (2020). Long-Term Face Tracking for Crowded Video-Surveillance Scenarios. *IJCB 2020*. Best student paper award (runner-up).

Barquero, G., et al. (2020). RimNet: A deep 3D multimodal MRI architecture for paramagnetic rim lesion assessment in multiple sclerosis. *NeuroImage: Clinical*.

TALKS

2024/12. DLBCN (Barcelona). FlowMDM: Seamless Human Motion Composition with Blended Positional Encodings.

2024/10. Google (Zurich). Inductive Biases in Motion: Human Motion Priors and Diffusion Models.

2024/06. Meta Reality Labs (Zurich). FlowMDM: Seamless Human Motion Composition with Blended Positional Encodings.

2023/12. DLBCN (Barcelona). BeLFusion: Latent Diffusion for Behavior-Driven Human Motion Prediction.

TEACHING

2024 - 2026. Flows and Diffusion Models, and Video Generative Models (Master in Computer Vision, UAB). **2022.** Advanced Algorithms (Bachelor in Computer Science, UB).

SUPERVISION

Master's thesis

Pablo Ruiz Ponce (2023). Text-Driven Multi-Human Motion Generation. *Universitat de Barcelona*.

Àlex Pujol Vidal (2023). Motion Binary Latent Diffusion. *Universitat de Barcelona*.

Johnny Núñez Cano (2022). Comparison of Spatio-Temporal Hand Pose Denoising Models. *Universitat de Barcelona*.

Nikita Belouossov (2022). Generating Sign Language Videos using DDPM. *Universitat de Barcelona*.

REVIEWING

Conferences.

CVPR (**2024-2026**), ECCV (**2024, 2026**), ICCV (**2025**), ACM Multimedia (**2023**), WACV (**2022**), BMVC (**2022**)

Journals.

International Journal of Computer Vision (**2023, 2024**)

International Journal of Social Robotics (**2023, 2024**)

Transactions on Visualization and Computer Graphics (**2022**)

Machine Learning (**2022**)